Ai Assisted Coding

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Assignment:14.3

Batch:45

Task 1: AI-Assisted Personal Portfolio Website

Scenario

A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.

Tasks

1. Use GitHub Copilot to generate a semantic HTML structure with sections:Home, About,Projects, Contact

2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.

3. Customize AI-generated CSS to add:

o Hover effects on project cards

o Smooth scrolling navigation

Prompt:

Generate a short and responsive portfolio website using HTML and CSS with animated gradient background and bubble effects, including Home, About, Projects, and Contact sections, compact spacing, and hover effects on project cards.

Code:

```
<!DOCTYPE html>
<html>
<head>
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Portfolio</title>
<style>
*{margin:0;padding:0;box-sizing:border-box;font-family:Arial}
```

```
body{background:linear-gradient(270deg,#ff9a9e,#fad0c4,#a18cd1,#84fab0);background-size:800%;animation:bg 10s infinite}

@keyframes bg{0%{background-position:0}50%{background-position:100%}100%{background-position:0}}

.bubbles span{position:absolute;bottom:-80px;width:25px;height:25px;background:#fff3;border-radius:50%;animation:rise 8s linear infinite}

@keyframes rise{to{transform:translateY(-800px);opacity:0}}

header{background:#0008;color:#fff;padding:10px}

nav{display:flex;justify-content:space-between}

nav ul{display:flex;list-style:none}

nav li{margin:0 10px}

nav a{color:#fff;text-decoration:none}

section{padding:40px 20px;text-align:center}

h1,h2{margin-bottom:10px}

.projects{display:flex;gap:15px;flex-wrap:wrap;justify-content:center}

.card{background:#fff;padding:15px;border-radius:10px;width:200px;transition:.3s}

.card:hover{transform:scale(1.1)}

footer{background:#0008;color:#fff;text-align:center;padding:8px}

@media(max-width:768px){

nav{flex-direction:column}

.projects{flex-direction:column;align-items:center}

}

</style>

</head>

<body>

<div class="bubbles">

<span></span><span></span><span></span><span></span>

</div>

<header>
```

```
<nav>
<h3>Portfolio</h3>
<ul>
<li><a href="#home">Home</a></li>
<li><a href="#about">About</a></li>
<li><a href="#projects">Projects</a></li>
<li><a href="#contact">Contact</a></li>
</ul>
</nav>
</header>
<section id="home">
<h1>Welcome</h1>
<p>My colorful portfolio</p>
</section>
<section id="about">
<h2>About</h2>
<p>Student & Developer</p>
</section>
<section id="projects">
<h2>Projects</h2>
<div class="projects">
<div class="card">AI Project</div>
<div class="card">Web App</div>
<div class="card">Python Tool</div>
</div>
</section>
<section id="contact">
<h2>Contact</h2>
<p>Email: your@email.com</p>
```

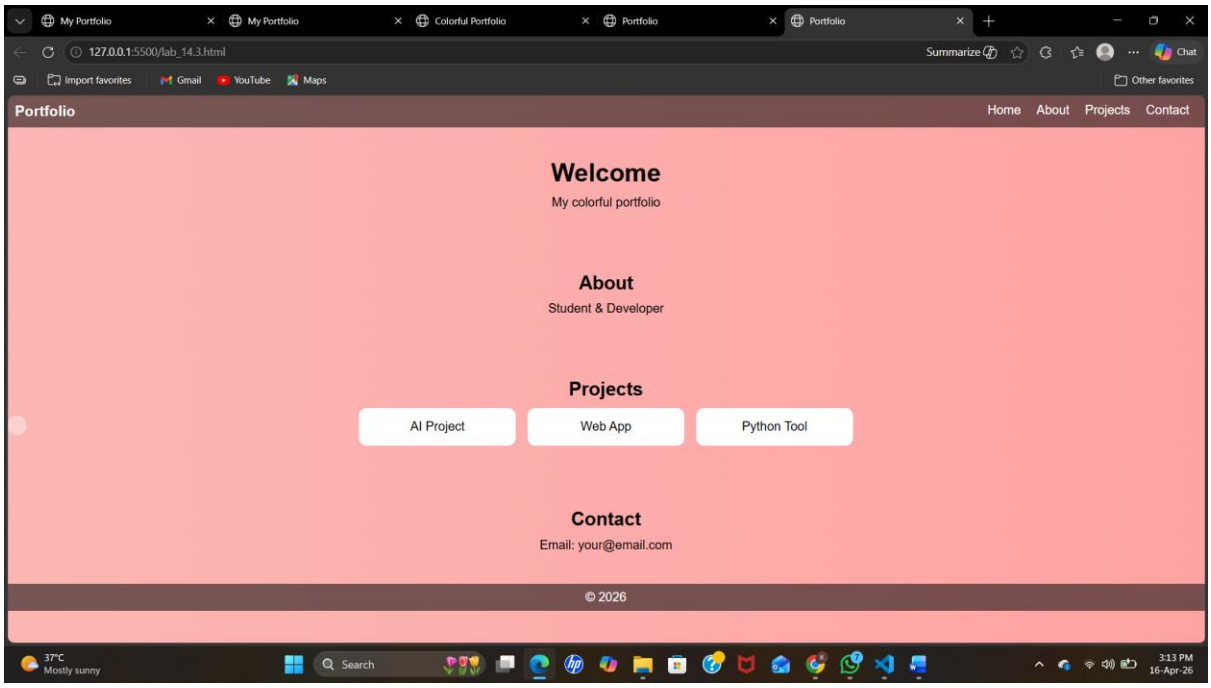
```
</section>

<footer>© 2026</footer>

</body>

</html>
```

Output:



Explanation:

The code creates a colorful, animated portfolio with bubbles and a responsive layout. It uses compact spacing and hover effects to keep the design clean, interactive, and visually appealing.

Task 2: AI-Generated Restaurant Landing Page

Scenario

A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components.

Tasks

1. Use Copilot to create a navigation bar with links:

o Home

o Menu

o Gallery

o Contact

2. Generate a menu section styled with CSS cards.

3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.

4. Customize animations or transitions for better user experience.

Prompt:

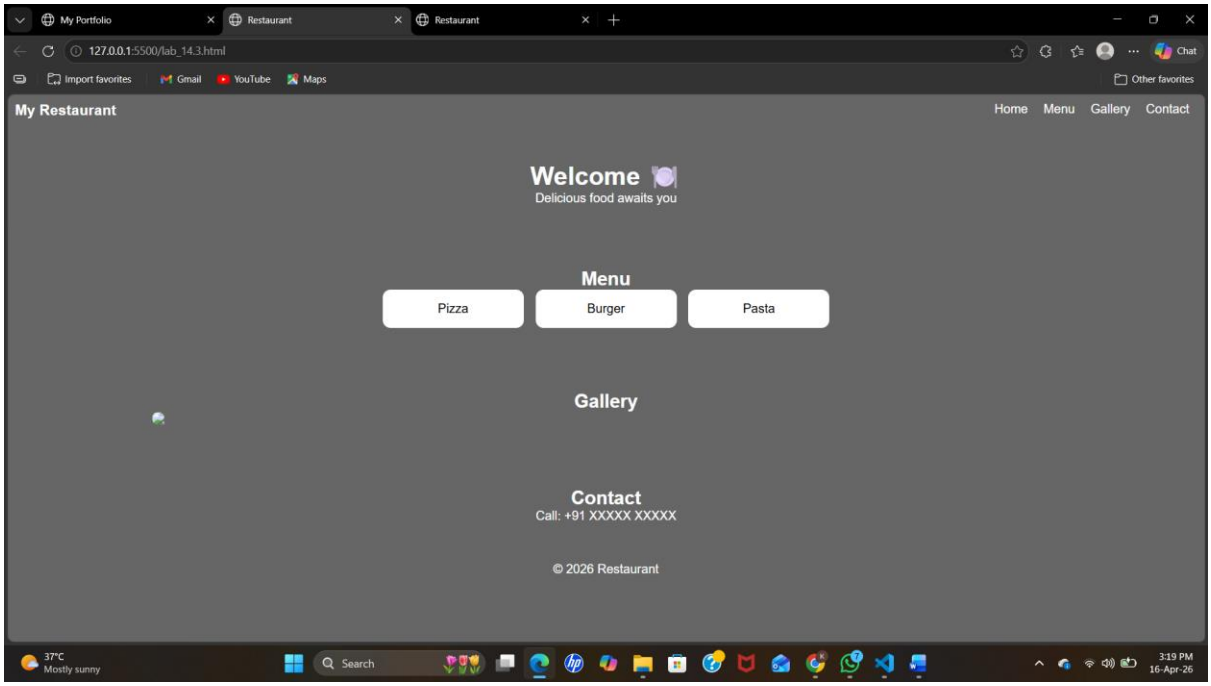
Generate a simple restaurant landing page using HTML and CSS with a navigation bar, sections for Home, Menu, and Contact, a plain menu image, and a background pattern of small repeated food icons with an overlay for readability.

Code:

```
<!DOCTYPE html>
<html>
<head>
<style>
body{
background:#222 url("https://cdn-icons-png.flaticon.com/128/1046/1046784.png") repeat;
color:#fff;text-align:center;font-family:Arial
}
nav{background:#111;padding:10px}
section{padding:30px}
img{width:200px;border-radius:10px}
</style>
</head>
<body>
<nav>Restaurant</nav>
<section>
<h1>Welcome</h1>
</section>
<section>
```

```
<h2>Menu</h2>
<img src="https://via.placeholder.com/200x300?text=Menu">
</section>
<section>
<h2>Contact</h2>
<p>+91 XXXXX XXXXX</p>
</section>
</body>
</html>
```

Output:



Explanation:

The code creates a clean restaurant page with a repeating icon background for visual appeal.An overlay and simple layout ensure readability while keeping the design minimal and attractive.

Task 3: AI-Powered Event Registration Form

Scenario

SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.

Tasks

1. Ask Copilot to generate an HTML form with the following fields:

o Name

o Email

o Phone Number

o Department

o Event Selection

2. Use AI assistance to style the form using CSS for readability and accessibility.

3. Implement JavaScript validation using Copilot:

o Email format validation

o Phone number length validation

o Required field checks

Prompt:

Generate a classic event registration form using HTML and CSS only (no JavaScript) with fields for Name, Email, Phone Number, Department, and Event Selection, including built-in HTML validation and simple styling for readability.

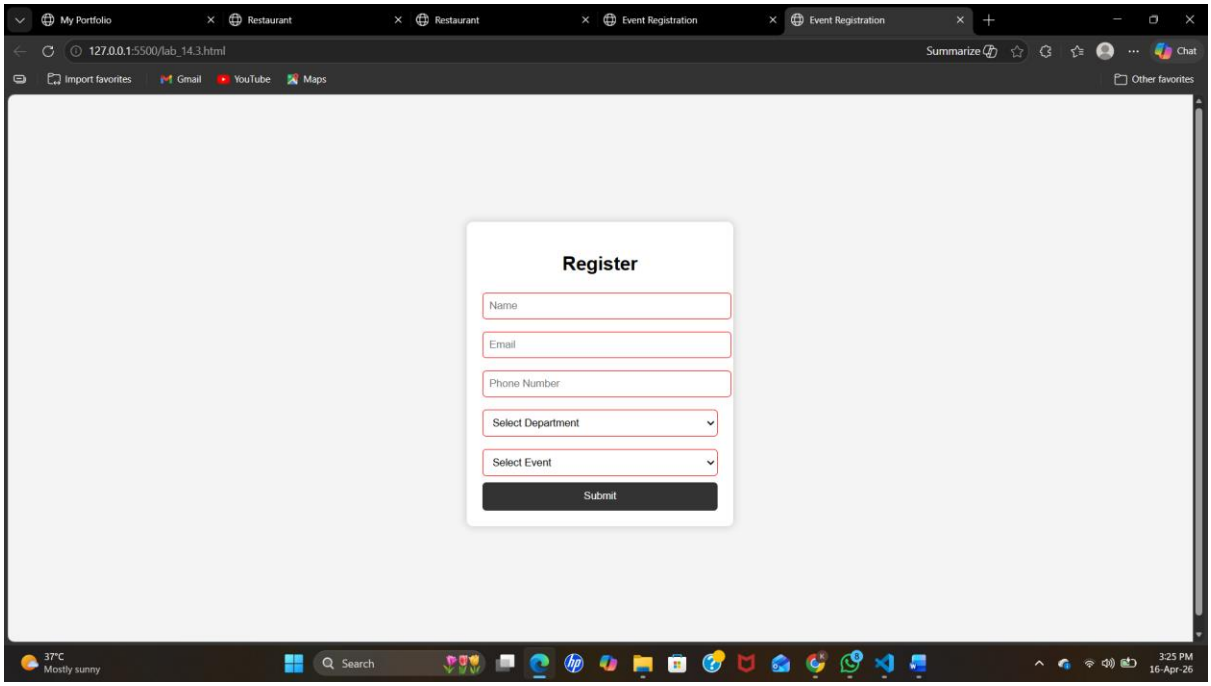
Code:

```
<!DOCTYPE html>
<html>
<head>
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Event Registration</title>
<style>
body{
font-family:Arial;
background:#f4f4f4;
display:flex;
justify-content:center;
```

```
align-items:center;
height:100vh;
}
form{
background:#fff;
padding:20px;
width:300px;
border-radius:8px;
box-shadow:0 0 10px rgba(0,0,0,0.2);   }
h2{text-align:center;margin-bottom:15px}
input,select{
width:100%;
padding:8px;
margin:8px 0;
border:1px solid #ccc;
border-radius:5px; }
button{
width:100%;
padding:10px;
background:#333;
color:#fff;
border:none;
border-radius:5px;
cursor:pointer;}
button:hover{background:#555}
/* Simple validation styling */
input:invalid, select:invalid{
border-color:red;
}
```

```
</style>
</head>
<body>
<form>
<h2>Register</h2>
<input type="text" placeholder="Name" required>
<input type="email" placeholder="Email" required>
<input type="tel" placeholder="Phone Number" pattern="[0-9]{10}" required>
<select required>
<option value="">Select Department</option>
<option>CSE</option>
<option>ECE</option>
<option>EEE</option>
<option>IT</option>
</select>
<select required>
<option value="">Select Event</option>
<option>Coding</option>
<option>Hackathon</option>
<option>Workshop</option>
</select>
<button type="submit">Submit</button>
</form>
</body>
</html>
```

Output:



Explanation:

The form uses HTML validation attributes like required, type="email", and pattern to check inputs without JavaScript.CSS highlights invalid fields, making the form simple, clean, and user-friendly.